

Parking Estimator

Test Cases, Equivalence Partitioning Tests, Boundary Value Analysis, and State Transition testing

Test Cases

Test Case 1 - Selection Long-Term Garage Parking

Priority Level 1 – High

My Assumptions Made for This Test Case

- \$2.00 per hour.
- \$12.00 daily maximum.
- \$72.00 per week (7th day free).

Purpose

The purpose of this test is to verify that Long-Term Garage parking selection using valid inputs will produce an output of the correct duration and cost estimate.

Testing Steps

- 1) Select Long-Term Garage parking from the section selector.
- 2) Enter an entry date of 03/01/2026.
- 3) Enter an entry time of 08:00am.
- 4) Enter a departure date of 03/01/2026.
- 5) Enter a departure time of 01:10pm.
- 6) Record the total parking cost estimate displayed. Result (A)
- 7) Change the departure date to 03/03/2026
- 8) Enter a departure time of 08:00am.
- 9) Record the total parking cost estimate displayed. Result (B)
- 10) Change the departure date to 03/07/2026
- 11) Record the total parking cost estimate displayed. Result (C)

Expected Results

- A) The total estimated parking cost should be \$12.00 even though the total time duration is only 5 hours and 10 minutes.
- B) The total estimated parking cost displayed should be \$24.00 for 2 full days.
- C) The total estimated parking cost displayed should be \$72.00 for 7 full days with the 7th day being free.

Actual Results

- A) \$12.00
- B) \$24.00
- C) \$72.00

Test Case 2 – Required Inputs

Priority Level 1 – High

A blank or unselected required input field should not allow a cost estimate to be calculated. If it does still produce a calculation or output parking cost estimate total then there is an issue with the logic of the triggers which cause the inputs to be calculated.

My Assumptions Made for This Test Case

- \$2.00 for the first hour.
- \$1.00 for each additional 1/2 hour.
- Input missing from required fields will result in no total cost being displayed.
- There will be an indicator to show the user which required fields are missing.

Purpose

The purpose of this test is to verify that a total cost will not be displayed unless the required fields contain user selected/input data.

Testing Steps

- 1) Select Short-Term parking from the section selector.
- 2) Enter an entry date of 03/01/2026.
- 3) Enter an entry time of 08:00am.
- 4) Leave departure date blank.
- 5) Leave departure time blank.
- 6) Record the total parking cost estimate displayed. Result (A)
- 7) Change the departure date to 03/01/2026.
- 8) Change the departure time to 9:30am.
- 9) Record the total parking cost estimate displayed. Result (B)

Expected Results

- A) The total estimated parking cost should be blank and there should be an indicator that you are missing the departure info input.
- B) The total estimated parking cost should now display the total of \$3.00 for 1 hour and 30 minutes.

Actual Results

- D) Invalid I assume but I am uncertain because it navigates away from the page to a blank screen.
- E) \$3.00

Test Case 3 – A Departure Date/Time Which is Earlier Than the Entry Date/Time is Prevented

Priority Level 1 – High

A departure date/time which is before the entry date/time should not be possible. This would result in a calculated total cost which is a negative amount.

My Assumptions Made for This Test Case

- \$18 is the cost per day.
- \$12 is the cost for five hours or less.
- The entry date/time must be before the departure time/date.
- There will be an indicator to tell the user that the departure date/time are invalid if they have entered a departure which is before the entry.

Purpose

The purpose of this test is to verify that an invalid departure date/time is detected and that calculating of the total duration is not allowed to occur until the date/time inputs are valid.

Test Steps

- 1) Select Valet Parking.
- 2) Enter an entry date of 03/02/2026.
- 3) Enter an entry time of 12:00pm.
- 4) Enter a departure date of 03/02/2026.
- 5) Enter a departure time of 08:00am.
- 6) Record the results taking note of if any invalid input message is displayed. Result (A)
- 7) Change the entry date to 03/01/2026.
- 8) Enter a departure time of 12:00pm.

- 9) Record the results taking note of if an invalid input message is displayed. Result (B)
- 10) Change the departure date to 03/3/2026.
- 11) Record the results verifying that no invalid input message is displayed. Result (C)

Expected Results

- A) The departure time input should be considered invalid, and an invalid input message should be displayed.
- B) The entry date input should be considered invalid, and an invalid input message should be displayed.
- C) There should no longer be an invalid input message, and the estimated parking cost total should be displayed as \$18.00 for 1 day.

Actual Results

- F) Invalid. No calculated total and the time is listed as negative 1 day
- G) Invalid. That actually gave an error message of "ERROR! Your Leaving Date Or Time Is Before Your Starting Date or Time"
- H) \$18.00

Test Case 4 - Parking Free 7th day is Correctly Applied

Priority Level 1 – High

The discount being incorrectly applied or not being applied to the total parking cost at all will result in inaccurate information.

My Assumptions Made for This Test Case

- \$2.00 is the cost per hour.
- \$9.00 is the daily maximum charge.
- \$54.00 is the charge per week (7th day free).

Purpose

The purpose of this test is to verify that the free 7th day for economy parking is applied correctly after the 6th full day and not before it.

Test Steps

- 1) Select Economy parking from the section selector.
- 2) Enter an entry date of 03/01/2026.

- 3) Enter an entry time of 08:00am.
- 4) Enter a departure date of 03/06/2026.
- 5) Enter a departure time of 12:00pm.
- 6) Record the total parking cost estimate displayed. Result (A)
- 7) Change the departure date to 03/07/2026
- 8) Record the total parking cost estimate displayed. Result (B)
- 9) Change the departure date to 03/08/2026
- 10) Enter a departure time of 08:00am.
- 11) Record the total parking cost estimate displayed. Result (C)
- 12) Change the departure date to 02/09/2026
- 13) Record the total parking cost estimate displayed. Result (D)

Expected Results

- A) The total estimated parking cost displayed should be \$53.00 for a total parking duration of 5 days and 4 hours.
- B) The total estimated parking cost displayed should be \$54.00 for a total parking duration of 6 days.
- C) The total estimated parking cost displayed should still be \$54.00 for a total parking duration of 7 days because the cost of 7th day will have been added and then subtracted from the total.
- D) The total estimated parking cost displayed should still be \$63.00 for a total parking duration of 8 days because the cost of 7th day will have been added and then subtracted from the total and then the 8th day will be charged.

Actual Results

- A) \$53.00
- B) \$54.00
- C) \$54.00
- D) \$63.00

Test Case 5 - Switching From Economy Parking to Valet Removes The Free 7th Day Discount

Priority Level 2 – Medium

The discount not being removed correctly from the estimated parking cost total will result in inaccurate information however it is in the favor of the patron so while it is important that it works correctly it is of less priority than if they were being over charged. It

doesn't stop the application from being used it only gives a discount when they are not entitled to one.

My Assumptions Made for This Test Case

- Economy parking is \$2.00 per hour.
- Economy parking is \$9.00 maximum cost per day.
- Economy parking is \$54.00 per week (7th day free).
- Valet parking costs \$18 per day.
- Valet parking is a \$12 cost for five hours or less.
- Changing the parking section selection will trigger the recalculation of the estimated parking cost total.
- The parking discount does not apply to valet parking.
- The parking duration will not reset when the parking section selection is changed.
- All discount indicators are removed from the viewable screen once the discount has been removed.

Purpose

The purpose of this test is to verify that the free 7th day for economy parking is removed if the section selection is changed to another type of parking section.

Test Steps

1. Select Economy parking from the section selector.
2. Enter an entry date of 03/01/2026.
3. Enter an entry time of 08:00am.
4. Enter a departure date of 03/08/2026.
5. Enter a departure time of 08:00am.
6. Record the total parking cost estimate displayed and make note of any indicators visible on the screen that a discount was applied. (A)
7. Change the parking section selection to Valet parking.
8. Record the total parking cost estimate displayed and verify that all discount indicators have been removed from the screen. Result (B)

Expected Results

- A) The total estimated parking cost displayed should be \$54.00 for a total parking duration of 7 days and there should be an indicator showing that a discount has been applied to the estimated parking cost total.

- B) The total estimated parking cost displayed should be \$126.00 for Valet parking with a total parking duration of 7 days and all indicators that a discount has been added should have been removed from the screen.

Actual Results

- E) \$54.00
F) \$126.00

Equivalence Partitioning Tests

Type of Parking	Parking Duration	Is Valid?	Expected Results	Actual Results
All Parking Sections	0 Hours	No	Invalid Duration	Valid for short term and valet parking the others were invalid
Short-Term	1 – 5.5 Hours	Yes	Valid Duration (Charged the Hourly Rate for Each Hour and then the 30 minute rate)	Valid
Short-Term	24 Hours	Yes	Valid Duration (Charged the daily rate and then the hourly or 30 minute rate accordingly)	Valid
Economy	1 - 5 Hours	Yes	Valid Duration (Charged the hourly rate)	Valid
Economy	24 – 124 Hours	Yes	Valid Duration (Charged the daily rate plus the hourly rate for 4 additional hours) (No 7 th Day Bonus)	Valid

Economy	125 To 168 Hours	Yes	Valid Duration (Charged the 24 hour rate for 7 days minus the bonus day if into the 7th day) (7th Day Bonus Applied)	Valid
Long-Term Surface	1 - 5 Hours	Yes	Valid Duration (Charged the hourly rate)	Valid
Long-Term Surface	24 – 124 Hours	Yes	Valid Duration (Charged the daily rate plus the hourly rate for 4 additional hours) (No 7th Day Bonus)	Valid
Long-Term Surface	125 To 168 Hours	Yes	Valid Duration (Charged the 24 hour rate for 7 days minus the bonus day if into the 7th day) (7th Day Bonus Applied)	Valid
Long-Term Garage	1 - 5 Hours	Yes	Valid Duration (Charged the hourly rate)	Valid
Long-Term Garage	24 – 124 Hours	Yes	Valid Duration (Charged the daily rate plus the hourly rate for 4 additional hours) (No 7th Day Bonus)	Valid

Long-Term Garage	125 To 168 Hours	Yes	Valid Duration (Charged the 24 hour rate for 7 days minus the bonus day if into the 7th day) (7th Day Bonus Applied)	Valid
Valet	1 minute - 5 Hours	Yes	Valid Duration (Charged the 5 hour of less rate)	Valid
Valet	5 or more	Yes	Valid Duration (Charged the 24 Hour Rate for each day of more than 5 hour and the 5 hour rate for each day less than the 5 hours)	Valid

Boundary Value Analysis

Short-Term Parking

Short-term Parking Section		
Expect Results		
Invalid	Valid	Valid
0 Minutes	1 minute to 168 hours	168 Hours and 1 minute
Actual Results		
Valid and charge the 1 hour rate of \$2.00	Valid at both ends of the range charging \$2.00 and \$168.00	Valid charged \$169.00 for the weekly plus 30 minutes

Decision Table Testing

Conditions	
Condition 1	The selected section is short-term parking
Condition 2	The selected section is economy parking
Condition 3	The selected section is long-term surface parking
Condition 4	The selected section is long-term garage parking
Condition 5	The selected section is valet parking
Condition 6	The parking duration is less than 5 hours
Condition 7	The parking duration is 5 hours or more
Condition 8	The economy/ long-term section parking duration is 168 hours or more

Actions	
Action 1	Calculate the half hour parking rate if selection is short-term parking
Action 2	Calculate the hourly parking rate if the duration is under 5 hours
Action 3	Round the duration to the daily parking rate if more than 5 hours
Action 4	Charge the daily 5 hour minimum rate if selection is Valet and the duration is less than 5 hours
Action 5	Apply the economy section 7 th day bonus

Rules				
Rule	Section	Hours	Bonus	Actions
Rule 1	Short-term	Rounded to next half hour if more than 1 hour and under 5 hours	No	Action 1 and Action 2
Rule 2	Economy	Less than 4 hours is charged at hourly rate	No	Action 2

Rule 3	Long-term Surface	Less than 4 hours is charged at hourly rate	No	Action 2
Rule 4	Long-term Garage	Less than 4 hours is charged at hourly rate	No	Action 2
Rule 5	Valet	Less than 5 hours is rounded to the 5 hour rate more than 5 hours is rounded to the day rate	No	Action 4
Rule 6	Long-Term Surface	168 hours is charged at weekly rate minus the 7 th day	Yes	Action 3 and Action 5
Rule 7	Long-Term Garage	168 hours is charged at weekly rate minus the 7th day	Yes	Action 3 and Action 5
Rule 8	Economy	168 hours is charged at weekly rate minus the 7th day	Yes	Action 3 and Action 5

Conditions	Rules							
	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8
Condition 1	Yes	No	No	No	No	No	No	No
Condition 2	No	Yes	No	No	No	No	No	Yes
Condition 3	No	No	Yes	No	No	No	No	No
Condition 4	No	No	No	Yes	No	No	No	No
Condition 5	No	No	No	No	Yes	No	No	No

Condition 6	Yes	Yes	Yes	Yes	Yes	No	No	No
Condition 7	Yes	No	No	No	Yes	Yes	Yes	Yes
Condition 8	No	No	No	No	No	Yes	Yes	Yes

Actions	Rules							
	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6	Rule 7	Rule 8
Action 1	Yes	No	No	No	No	No	No	No
Action 2	Yes	Yes	Yes	Yes	No	No	No	No
Action 3	No	No	No	No	No	Yes	Yes	Yes
Action 4	No	No	No	No	Yes	No	No	No
Action 5	No	No	No	No	No	Yes	Yes	Yes

State Transition Testing

States Explained	
State	Definition
State 0	Application is opened
State 1	Parking section is selected
State 2	Parking duration is entered
State 3	Parking duration and parking cost is rounded to the 24-hour rate
State 4	Parking total cost is displayed
State 5	Parking total cost is displayed with the 7 th day bonus applied

State Transition Test 1

Test User Input Values

- Parking Section: Short-term
- Parking Duration: 4 and half hours

Path

State 0 > State 1 > State 2 > State 4

Expected Results

\$9.00 should be the total parking cost.

Actual Results

\$9.00 should be the total parking cost.

State Transition Test 2

Test User Input Values

- Parking Section: Economy
- Parking Duration: 25 hours

Path

State 0 > State 1 > State 2 > State 3 > State 4

Expected Results

\$11.00 total parking cost.

Actual Results

\$11.00 total parking cost.

State Transition Test 3

Test User Input Values

- Parking Section: Long-term Garage
- Parking Duration: 27 hours

Path

State 0 > State 1 > State 2 > State 3 > State 4

Expected Results

\$18.00 total parking cost.

Actual Results

\$18.00 total parking cost.

State Transition Test 4

Test User Input Values

- Parking Section: Economy
- Parking Duration: 30 hours

Path

State 0 > State 1 > State 2 > State 3 > State 4

Expected Results

\$18.00 total parking cost.

Actual Results

\$18.00 total parking cost.

State Transition Test 5

Test User Input Values

- Parking Section: Economy
- Parking Duration: 168 hours

Path

State 0 > State 1 > State 2 > State 3 > State 5

Expected Results

\$54.00 total parking cost. With the free 7th day

Actual Results

\$54.00 total parking

Which testing techniques were most effective for you, and why?

I found errors in web app using the test cases and during equivalence partitioning as well as in boundary value analysis so I would say those three were the most effective for me at discovering errors. However, I don't think testing is just about finding errors, it is also about finding out what parts of the program are stable and valid, so we have a better idea of where our testing efforts are probably needed and what is probably in good shape. As the book authors have stated numerous times 100% testing is not possible so anything which

can direct our focus towards what needs our attention the most is very helpful. (Van Veenendaal et al., 2019a).

References

Van Veenendaal, E., Black, R., Graham, D., & Services, R. B. R. B. C. (2019a). Foundations of software testing: ISTQB Certification.